

INTRODUCTION



ReviewSense AI is an intelligent sentiment analysis system that analyzes Arabic and English product reviews using NLP and Transformer-based AI models.

The system classifies reviews into Positive, Negative, or Neutral sentiments, with support for Saudi dialect analysis to better understand local Arabic content.

ReviewSense AI helps businesses and users gain valuable insights from customer feedback and supports data-driven decision-making.

Objectives

- Analyze customer reviews automatically
- Support Arabic and English
- Classify sentiments (Positive / Negative / Neutral)
- Detect recurring issues in reviews
- Support Saudi dialect
- Provide data-driven insights

Methodology



Text Preprocessing

- Text Cleaning
- Tokenization
- Stopword Removal
- Text Normalization
- Lemmatization



Experiments & Results

- Multiple AI models were tested and evaluated using multilingual datasets
- Accuracy and classification metrics were applied to measure performance

The system achieved promising sentiment classification results for Arabic and English reviews, with improved support for Saudi dialect analysis.

The Ensemble Voting System enhanced prediction reliability across multiple AI models.

AI Models

The system integrates Transformer-based models including:

- AraBERTv2
- MARBERTv2
- XLM-RoBERTa

Technologies

- Python
- Flask
- React / Next.js
- PostgreSQL
- Docker
- Hugging Face



System Features

- Multilingual Sentiment Analysis
- Saudi Dialect Support
- Interactive Dashboard
- Topic Extraction
- CSV File Upload & Analysis
- Real-Time Analysis Results
- Ensemble Voting System for Improved Prediction Accuracy

CONCLUSION

ReviewSense AI provides an intelligent and efficient solution for analyzing large volumes of customer reviews using AI and NLP technologies.

The system enables businesses and users to better understand customer opinions, improve products and services, and support data-driven decision-making through automated sentiment analysis.

Prototype

